

Jeffery Ye

s.jeffery.ye@gmail.com | [linkedin.com/in/jefferyye](https://www.linkedin.com/in/jefferyye)

Software and AI engineer with experience developing and leading full-stack applications (TypeScript, Django) and agentic AI solutions (LangChain, PydanticAI) at scale.

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, R

AI/ML: Agentic AI, LangChain, PydanticAI, LLMs, RAG, NLP, Agent Driven Development

Full-Stack: React, Node, SQL (Postgres, MySQL), NoSQL (Neo4j, pgvector), Flask, FastAPI, TanStack

Platforms & DevOps: AWS, Spark, Databricks, GCP, Docker, Azure, Git, CI/CD, Linux/Unix, Hadoop

PROFESSIONAL EXPERIENCE

Software Engineer | U.S. Department of the Treasury (Tech Force) | June 2026 - Present

- Developing business-oriented solutions with JS/TS, TanStack, Terraform, AWS, PostgreSQL, pgvector, OIDC/Auth.

Software/AI Research Assistant | Seattle Children's Hospital | April 2025 - June 2026

- Designed, developed, and deployed two multi-agent AI systems to automate genomics research workflows, delivering tools that **reduced experimental design time by 95%**.
- Engineered a scalable data pipeline integrating SQL, graph (Neo4j), and vector (pgvector) databases to synthesize insights from unstructured literature, genomic, and proteomic data.
- Developed secure full-stack applications with Django, Flask, and FastAPI for researchers.

Teaching Assistant | University of Washington | March 2025 - June 2026

- Taught AI/ML, data science, LLMs, and AI Ethics to 200+ students over 2 years, achieving a **4.9/5.0 rating** (Top 20% among all UW instructors).

PROJECTS AND PUBLICATIONS

Automated extraction and optimization of protein purification protocols using multi-agent large language models by Jeffery Ye et al. | [bioRxiv](https://www.biorxiv.org/content/10.1101/2026.01.01.478123v1) | 2026

- Led the design, development, and evaluation of the full-stack application, coordinating with domain experts to iterate on designs and validate outputs.

AI Expense Categorizer | 2025

- Ideated and developed a fintech AI agent with PydanticAI to automate the secure classification of unstructured financial data, completing **~2 hours of manual work in 30 seconds**.

Musicalendar | 2025

- Developed a full-stack web application using ReactJS and a Flask back-end, deployed on AWS.
- Engineered a max-flow scheduling algorithm to optimize high-volume, 1-on-1 meeting bookings, replacing inefficient manual coordination via email chains and spreadsheets.

Educational Research STM | 2024

- Applied structural topic modeling to analyze **800+ educational policy research papers**, developing expertise in NLP pipelines, Spark, and statistical validation of topic model outputs.

Optum Claims Analysis | 2023

- Engineered SQL queries with JDBC to analyze 50k+ urinary cancer records in the Optum dataset.

EDUCATION

BS Informatics (Focus: Software & AI) | University of Washington | June 2026

- GPA:** 3.6